ADAMS assignment 2

a walk through the multiverse

In Assignment 1, you saw how questionable research practices may distort the scientific record. This assignment is centered around one possible answer to the flexibility that researchers typically have when processing and analyzing their data: multiverse analysis. aS in Assignment 1, we will get there in steps:

1. Come up with a single data-processing and analysis pipeline for a given research question and dataset.
2. Verify the outcome of another potential pipeline (co-piloting).
3. Come up with plausible alternative choices for data-processing and analysis decisions.
4. Combine these alternatives in a so-called multiverse of analysis pipelines.
5. Conduct those analyses (aka a multiverse analysis) and report the outcomes.

Practicalities

You will only be graded for the final, full report. The intermediate milestones have to be handed in (pass/fail), but they don't have to be perfect. Use them to ask questions and get feedback on your work.
If you run into any issues, please let the teachers know.

Part A: Single pathway analysis

Description

In this part, the goal is to answer a research question by processing and analyzing a given dataset (the dataset can be found on Brightspace). Below, you can find a description of the data and the two corresponding research questions. You don't have to come up with your own questions. Instead, it is important to pick an appropriate way to analyze the data to answer the given research questions. For now, it is sufficient to pick *one* approach to process and analyze the data. The dataset consists of several variables. Some may not be *directly* related to the research question, but might nevertheless be relevant in terms of processing the data or model building. After you have conducted the analyses, interpret the outcomes carefully in line with the pointers given in the lecture.

Study description

One of the main selling points of teaching statistics in study programs like psychology is that students become critical consumers of science and science communication. The current study sought to examine whether statistics proficiency indeed predicts someone's ability to spot mistakes in news articles describing scientific discoveries. Participants read 14 news articles, seven of which contained a mistake, the other seven were accurate. For each article, they had to determine whether it contained a mistake ("fake news") or not. In addition, they had to answer one question about the content of each article, to check whether participants were paying attention and not just guessing or responding randomly.

Research questions

- Is there a positive association between the number of credits obtained from statistics courses and their ability to detect fake news?
- To what extent does this relation vary depending on reading comprehension (= the attention check)?

Participants

237 people who recently obtained a degree in data science, AI, statistics or related fields.

Variables (name: variable measurement, range or levels)

- ID (participant identification number)
- Sexe: categorical, female (0)/male (1)
- Age (in years): numerical, 20 (youngest) – 36 (oldest)
- StatsCourses (number of stats courses passed, expressed in credits also referred to as ECs): numeric, 12 (lowest) – 94 (highest)
- GradeStats (average grade on the stats courses participants took): numeric, 5.0 (lowest) – 9.5 (highest)
- ReadNews (how often participants read the newspaper): numeric, ratings on a discrete 7-point scale from 1 never to 7 daily

- Judgement (after reading each article, participants had to indicate whether it contained a mistake or not; this variable indicates the number of correct decisions): numerical, 0 (lowest) – 14 (highest)

- Comprehension (reading comprehension check, after each article participants also answered a relatively simple two-alternative multiple choice question about the article): numerical, 5 (lowest) – 14 (highest)

- GoalStudy (during the debriefing, participants were asked what they thought the goal of the study was): categorical, 1 (guessed correctly)/ 0 (guessed incorrectly)

Hand in

We ask you to hand in two documents:

- A document describing the data processing and analysis pipeline as well as the outcomes of those analyses and how they answer the research questions. Write this down in essay format, of max. 500 words. If you have any questions for the teacher, you can add those at the end.

- Your script (.R or .py), with comments to explain every part of the data-processing and analysis pipeline.

Part B: Checking computational reproducibility

Description

You will receive a similar document to the one you handed in for Part A, though the data-processing and analysis choices will not be identical to yours. The idea is that you try to reconstruct the code for the data-processing and -analysis steps to see whether you can obtain the same outcomes and come to the same conclusions (you will only receive the report, not the code itself).

Hand in

We ask you to hand in two documents:

- A document describing the reproduced outcomes and to what extent they are similar or different compared to the original of max. 500 words.
- Your script (.R or .py), with comments to explain every part of the data-processing and analysis pipeline.

Part C: Constructing the multiverse

Description

As explained in the lecture of this week, and as you also experienced when completing Parts A and B of this assignment, there are often different plausible ways to process and analyze a given dataset when answering a particular research question. In a multiverse analysis, one seeks to make those (hidden) decisions explicit to examine how robust a finding is to plausible alternative data-processing and -analysis options. In other words, one makes an inventory of all decisions and corresponding alternatives and then combines them to form a multiverse of data-processing and analysis pathways. We will do the same here for our study involving the association between the number of credits obtained from statistics courses and someone's ability to detect fake news. We will do this in two stages though. This week, your assignment is to come up with the decisions and plausible alternatives *without* performing the analyses themselves. By comparing your single pathway analysis with the one you received in Part B, you may already discover some options. Note that in this part (and Part D), you will focus on the first research question, concerning the association between statistics credits and the ability to detect inaccurate news. You do not need to construct/conduct a separate multiverse for the second research question.

Hand in

A document describing different processing and/or analysis decisions and corresponding choices of max. 500 words. This doesn't have to be in essay form (yet); it can be a list with a short explanation about each decision and choice. You also don't have to perform any analyses yet. If it helps, feel free to include a visualization/drawing of the "garden of forking paths". At this stage, don't worry about including too many or wrong paths. The idea is that, based on the feedback, you will adjust, trim or expand the multiverse before conducting the actual analyses in the next part of the assignment.

Part D: Conducting the multiverse analysis

Description

The idea is to finalize the multiverse and subsequently perform the respective analyses. Note that not every technically possible analysis constitutes a defensible alternative. Your final multiverse should include choices that can be justified on substantive or methodological grounds. In addition, it is important to be mindful about the number of unique pathways in your multiverse. For example, if you identified four decisions with two options each, you'd get 2 x 2 x 2 x 2 = 16 different pipelines, each with their own outcome. This number can grow quickly. We therefore advise you to focus on justifiable options and not to come up with as many options as possible. Also, when writing the code, you could in theory perform every analysis separately, or you can be more efficient and write functions that you can reuse and/or use for-loops, for example. Once you obtained all the outcomes, use an appropriate way to summarize and visualize the results to come to a conclusion about how robust the effect in question is to different plausible data-processing and analysis decisions.

Hand in

We ask you to hand in two documents:

- A document describing different processing and/or analysis decisions and corresponding choices (revised version of what you handed in for Part C). In addition, the document should describe the outcomes of the multiverse analysis, including a visualization of the results, and how this relates to the robustness of the effect. See the lecture on multiverse analysis for guidance in this regard The document should be max. 1000 words.
- Your script (.R or .py), with comments to explain how the different decisions and choices of the multiverse are implemented.

Full report

Description

For the final deadline, update your report using the feedback you have received. Read everything once again from the top to make sure it all follows logically. The full report should first describe your single pathway analyses from Part A and its outcomes, then describe the alternative single pathway analysis from Part B and its outcomes, and subsequently describe the multiverse (i.e., rationale for the decisions and choices) and its outcomes, including a visualization of the results and an interpretation of the findings. Make sure to connect the two single pathway analyses (from Part A and B) to the multiverse; are the results similar to those from other plausible choices, or considerably different to some extent (this can be done descriptively). Also, reflect on what a multiverse is meant to accomplish and how it deals with the data-analytic flexibility that researchers typically have. You do not have to integrate Part B in this description, but rather include it as an Appendix (which will be graded as well). The maximum word count for the full report is 2000 words Some final checks:

- Does each team member support all the statements made in the report and understand all lines of code?
- Does your script run, from start to finish, in a clean session, and match the reported results exactly?
- Did you include the AI statement as well?

You will have a short oral exam about the assignment (more info on Brightspace). Together with the report you hand in, this is the basis for your assignment 2 grade.

Hand in

1. **The full report (pdf)**
2. **The full script (.R or .py)**